

Developing Enterprise Handheld Applications with SQL Anywhere Studio

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Sybase Inc.

3Com
More connected.

Mobile and Embedded Computing Mission

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Extending Corporate Information



- Delivery of enterprise information anywhere
- Synchronize information seamlessly ***from the enterprise to…***
  - Workgroups
  - Desktops
  - Laptops
  - Handheld computing devices
  - Pagers
  - Intelligent devices

…and back again

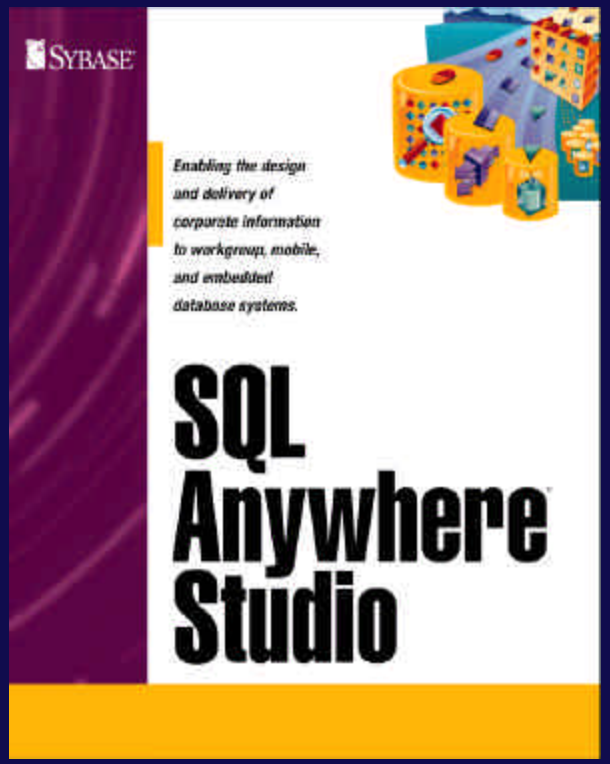
Mobile Database Industry Leader

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"Sybase is the clear leader
for the 3rd consecutive year with
55% marketshare in the
mobile database market."

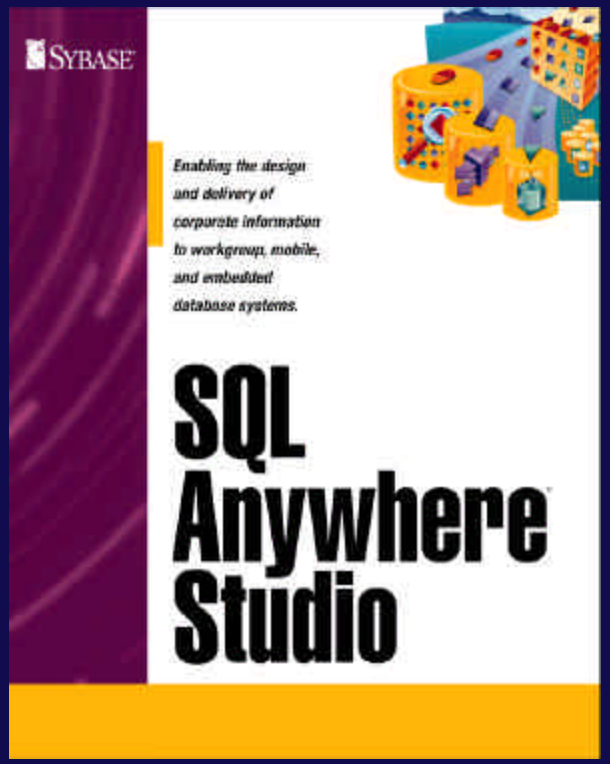
GartnerGroup's Dataquest,
Embedded Database Vendors Focus on Mobile Devices, March 1999

The Leader in Mobile Computing Experience



- Over *4,000,000* seats deployed
- Over *10,000* corporate customers
- Over *400* embedded partners

Sybase SQL Anywhere Studio



The Complete Solution:

- Leading mobile database
- Small fingerprint deployment
- Administration tools
- Data modeling and design
- Query and reporting
- Web enabled
- Update anywhere replication and synchronization
- Open interfaces

Common Device Characteristics

- Common characteristics
  - No hard disk
  - Limited memory (2MG)
  - Applications always stay running





- Development environment
  - C-based development environment
  - Program data management structures from scratch





Sybase and 3Com Strategic Alliance

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Enable enterprise data and applications for Palm Computing Platform

3Com®

SYBASE®

UltraLite Deployment Option

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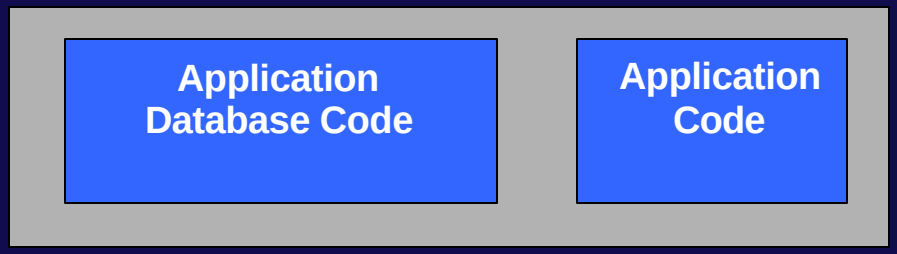
- Small fingerprint database
  - Application-optimized database technology
  - Analyzer determines required features and tables
  - Builds in-memory database (as small as 50K)
- High Performance
  - Optimized database for in-memory use
  - Support for indexes for fast data updates and retrievals
- Increased Developer Productivity
  - Use of industry standard SQL for data storage, retrieval and manipulation

UltraLite Development and Deployment

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- Step #1: Develop application code
- Step #2: Invoke UltraLite Analyzer
- Step #3: Compile and Link
- Step #4: Deploy
- Step #5: Setup Synchronization

Step #1 Develop Application Code

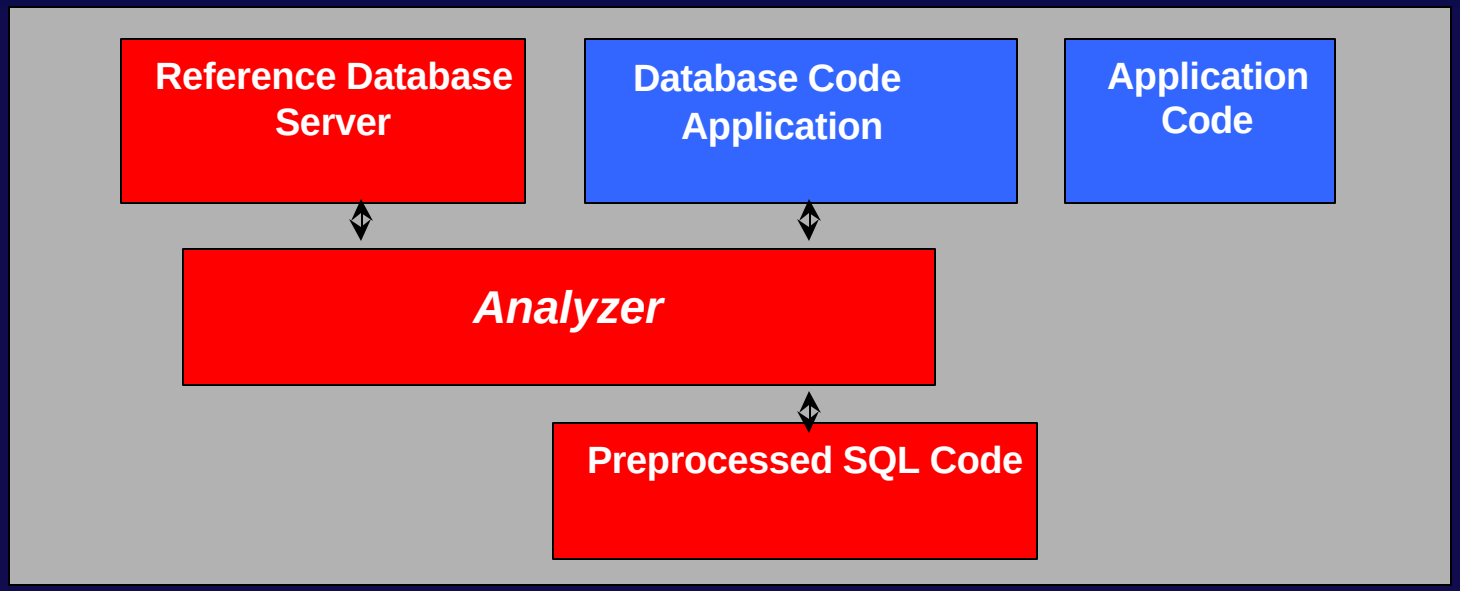


- Database access and GUI code
  - Contains static embedded SQL to define queries



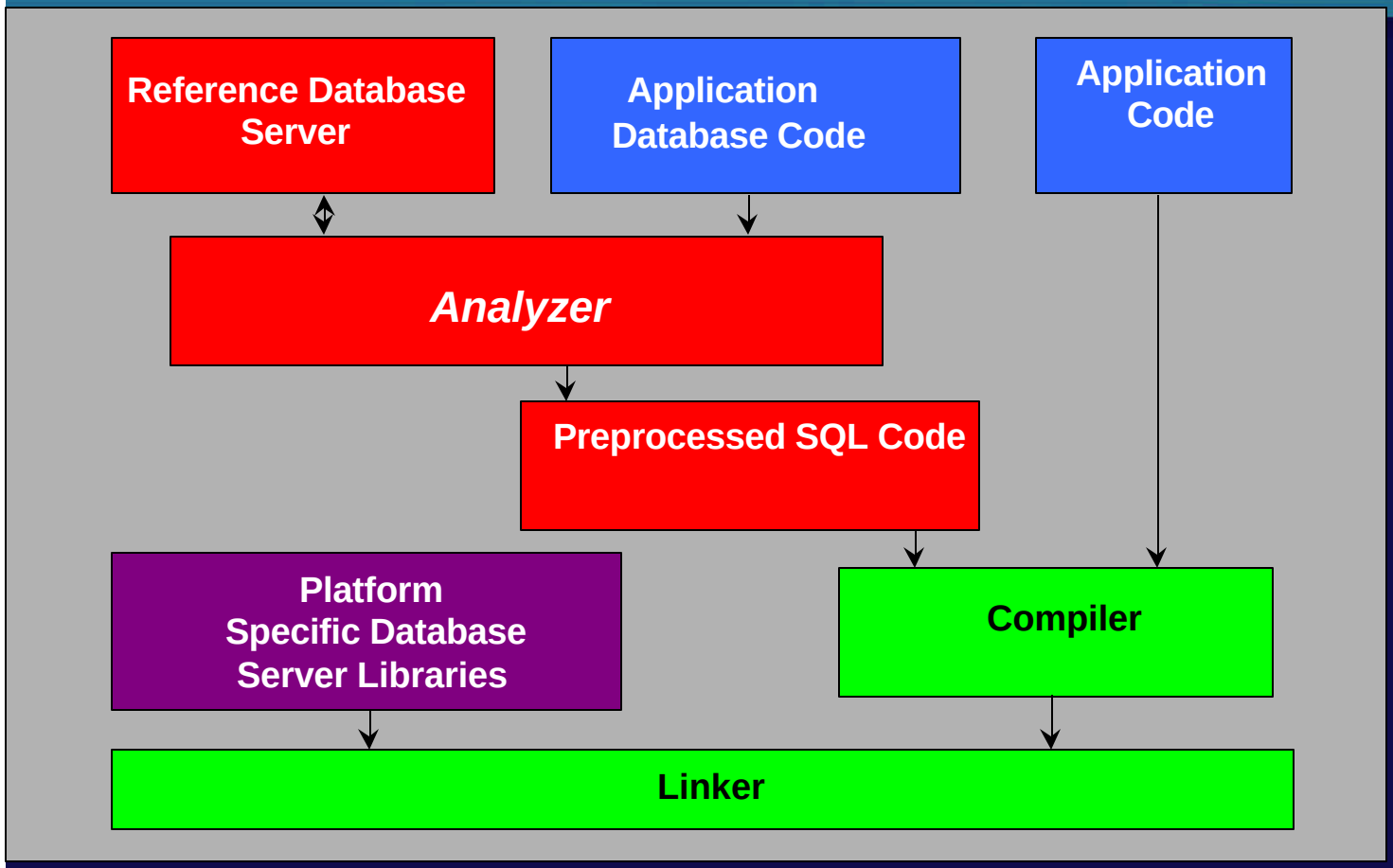
- Seamless integration with CodeWarrior

Step #2 Invoke UltraLite Analyzer



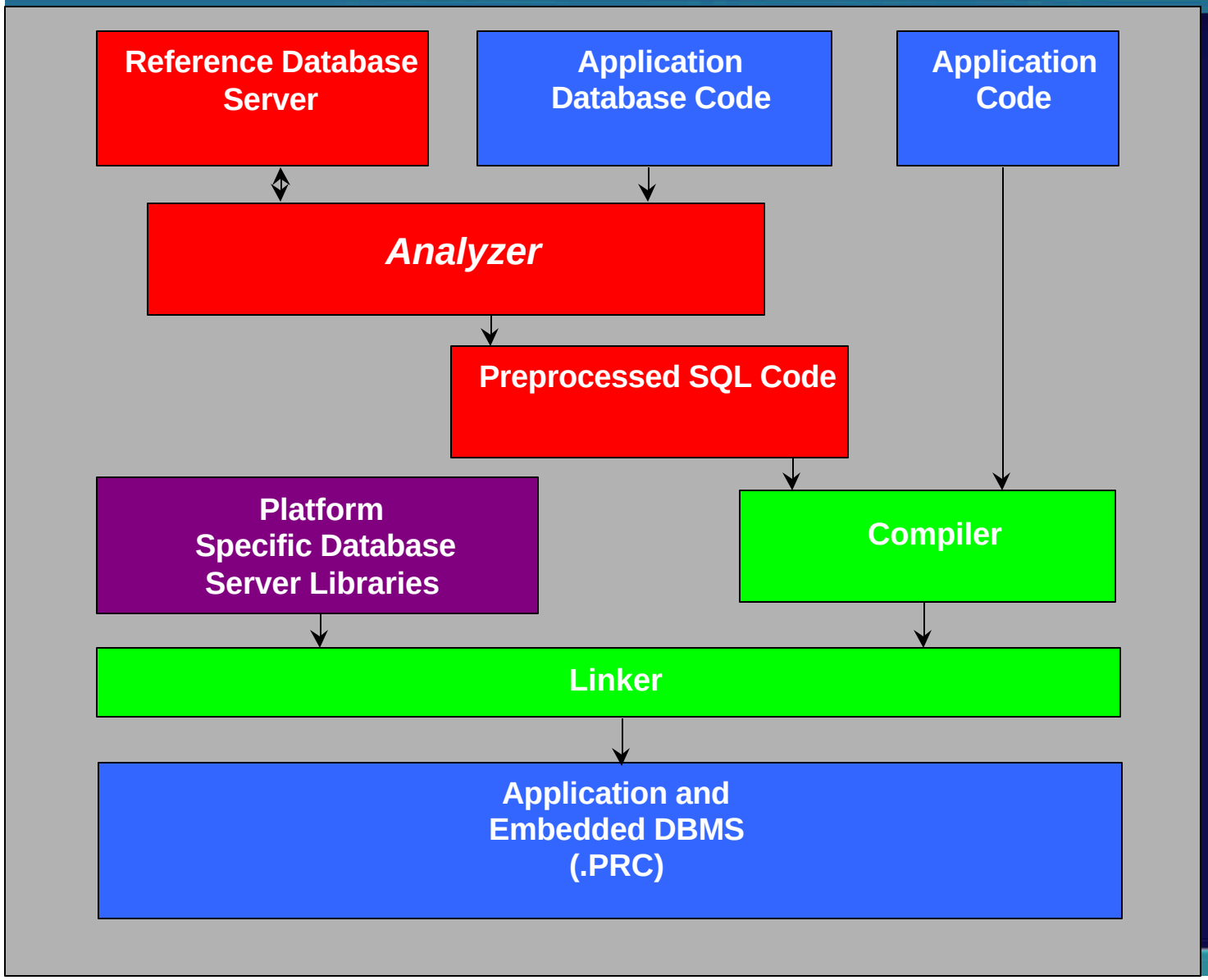
- Analyzer determines database schema
- Extracts required database pieces from SQL Anywhere
- Generates application-specific database

Step #3 Compile and Link



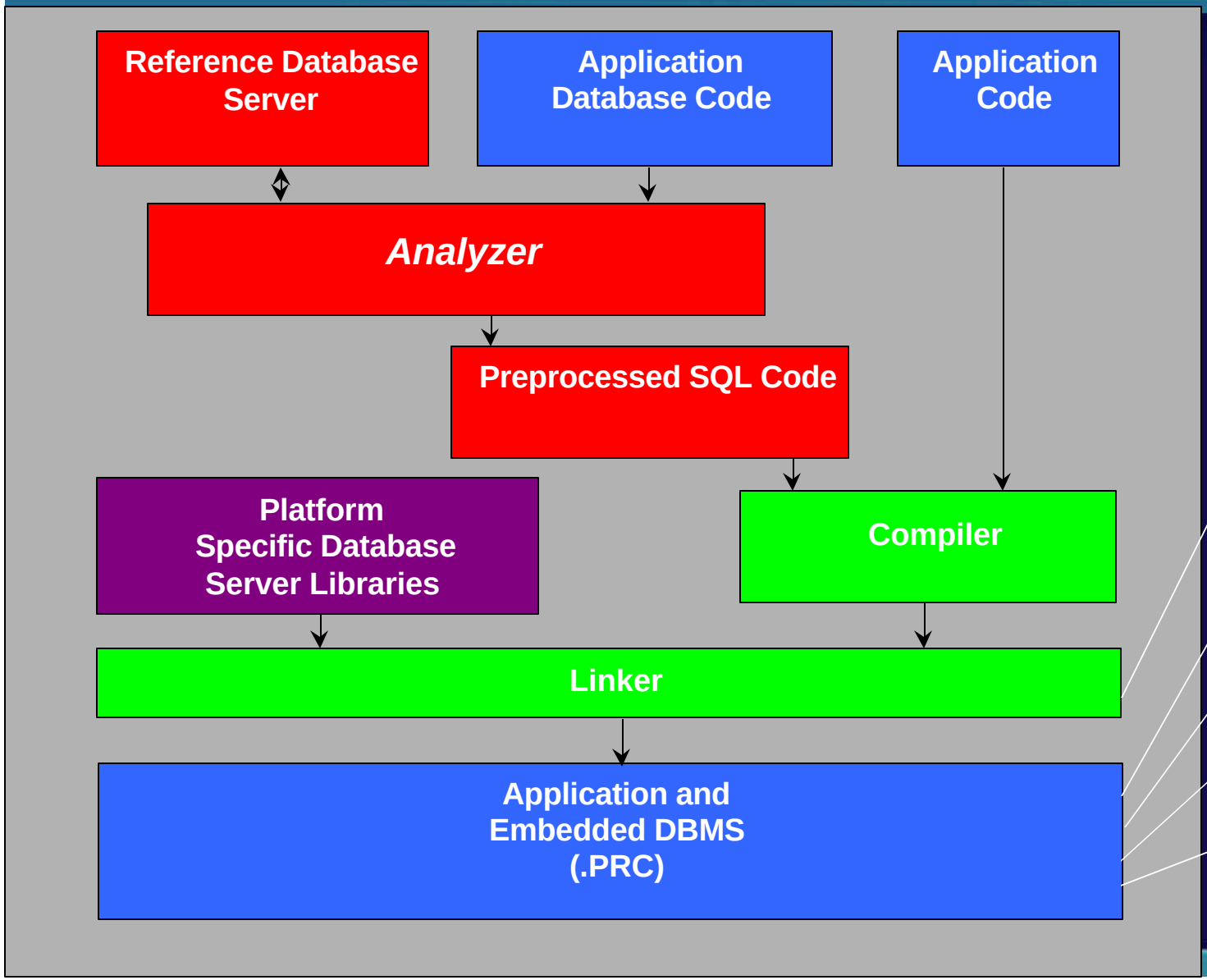
- Link in platform specific libraries for deployment platform

Step #4 Deploy



Step #4 Deploy

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UltraLite Database Technology Benefits

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- Rich functionality
  - Referential integrity
  - Complex queries, joins
  - Transaction processing
- Increased productivity
  - Develop to SQL API
- Improved performance
  - Indexes
  - In memory database
- Synchronization
  - APIs to communicate to MobiLink server

MobiLink Server Synchronization Technology

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- Optimized for *data exchange* between handheld devices and enterprise servers
- Two-way *session-based* synchronization to UltraLite database
  - TCP/IP, serial cable
- Direct synchronization via ODBC
  - Sybase, Oracle, Microsoft, IBM

Synchronization Components

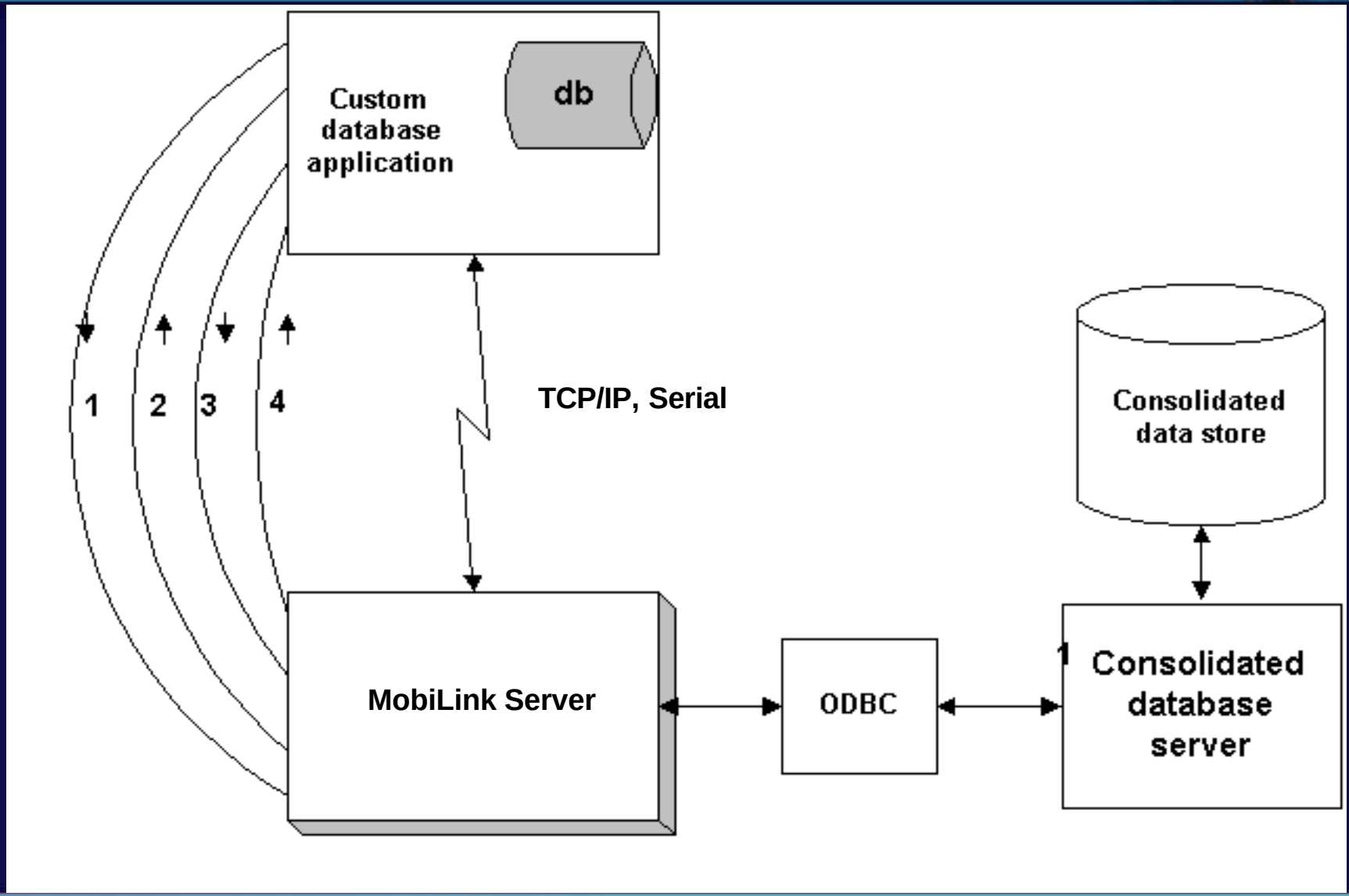
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- Synchronization logic
  - Automatically built into UltraLite application
  - UltraLite runtime keeps track of changes to data
- MobiLink Synchronization Server
  - Provides interface between consolidated database and UltraLite application
  - Works with any ODBC-based database
    - Sybase, Microsoft, Oracle, IBM

Two Main Synchronization Steps

- Upload
  - UltraLite keeps track of rows inserted, updated, deleted with state-bytes
  - UltraLite establishes and maintains connection with MobiLink server
  - For each table
    - Only records that have changed are uploaded
- Download
  - MobiLink uses synchronization scripts to retrieve changed data from consolidated database

The MobiLink Synchronization Process



Synchronization Scripts

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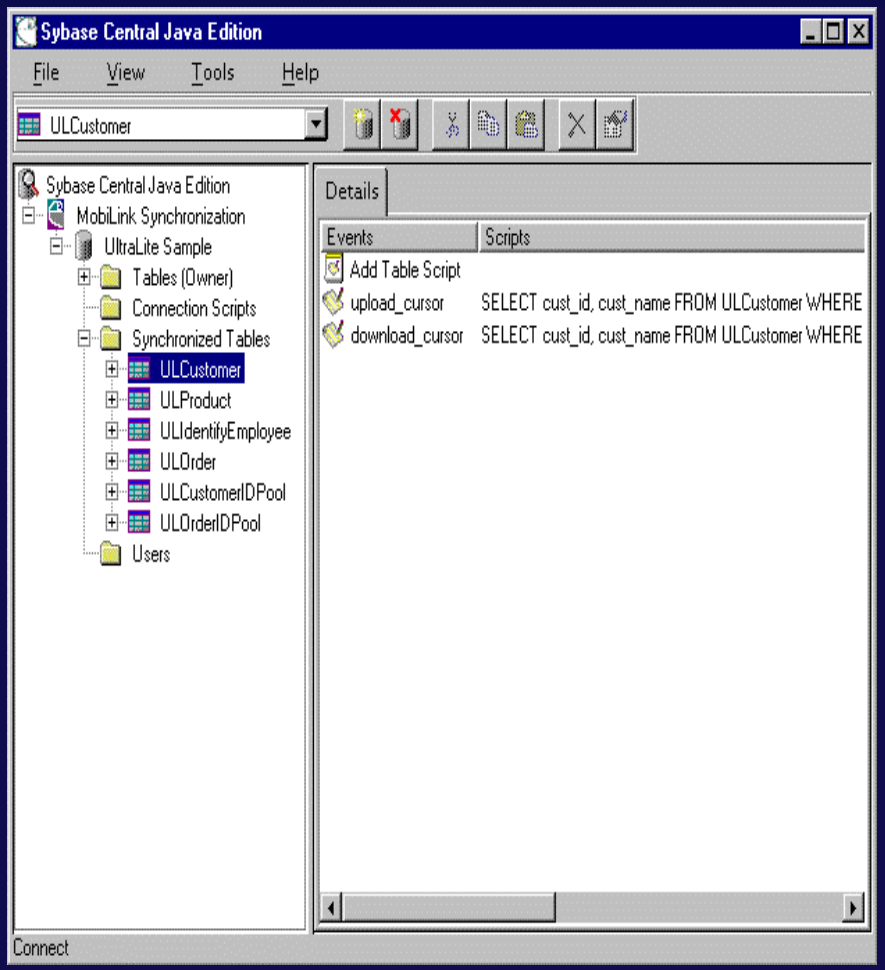
- SQL statements executed against the consolidated database
  - Written in language of consolidated database
- Guides the synchronization server
  - Handles connections
  - Controls the flow of data in both direction
  - Synchronization events
    - Connection level
    - Table level
- MobiLink provides default scripts

Conflict Detection

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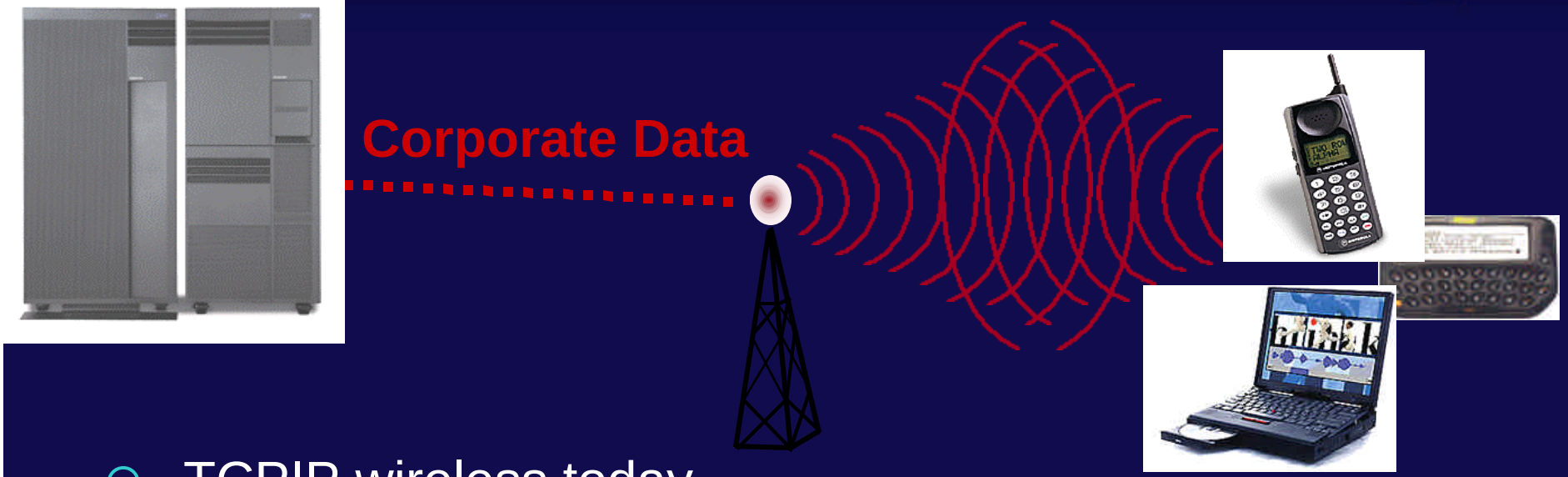
- Conflict arises during upload of rows to consolidated database
- UltraLite maintains old value and current value
  - MobiLink compares old values with consolidated database
  - If old values do not match there IS a conflict
- Conflict Resolution
  - Inserts old and new values into temporary tables
  - Calls resolve conflict script which can execute stored procedures or define resolution steps

Ease of Administration
The MobiLink Manager



- Graphical environment for centralized administration
  - Based on Java version of SQL central
  - Wizards to aid in creation of scripts
  - Aids in conflict resolution

Advanced Wireless Synchronization



- TCPIP wireless today
- Planned support for additional wireless protocols
  - Mobitex, CDPD, DataTAC
  - Prioritization of messages
- Planned support for PalmVII
- Built-in security
  - Encryption, guaranteed delivery

Partners and Alliances

































Customer Example
Field Force Automation

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- EZ-Data, Inc.
  - Locally store client history on Palm III, update and synchronize with consolidated database
  - Increased employee productivity and customer service





"We are pleased that Sybase is continuing to offer leading-edge technology in the mobile space. Many of our 75,000 users in the insurance industry have expressed interest in our Client Data System on Palm-based devices, and we look forward to working with Sybase and 3Com to spur our growth in the handheld marketplace."

Customer Example
Point of Care

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- Dymar Medical Solutions
  - Allows physicians to access and update critical patient information by scanning patient's id bracelet
  - Synchronize the patient information using MobiLink with various backend systems
  - Reduction in administration costs and increased patient profile accuracy

Dymar Medical Solutions Inc.

symbol

"Dymar was looking for a solution that enabled us to build industrial strength applications with the ability to grow and integrate with the enterprise. Sybase's UltraLite and MobiLink technologies provided the perfect fit by enabling us to leverage industry standard SQL to quickly develop our workload measurement application for the Palm Computing platform and seamlessly synchronize that data to a variety of corporate database systems."

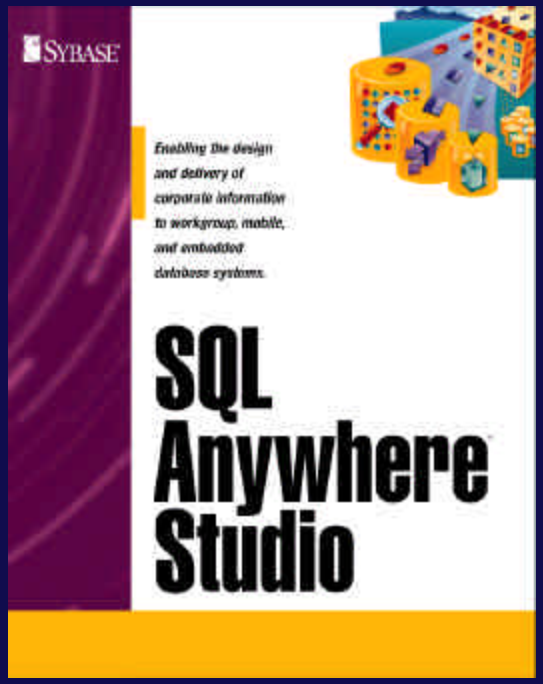
Comprehensive Programs

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- Business Solutions Alliance Program
  - Provides focused programs for OEM partners, platform vendors, system integrators
  - Offers aggressive pricing, priority technical support, marketing programs
- Sybase Developer Network: www.sybase.com/sdn
  - Provides developers with timely and convenient access to beta and release products, technical information, events, and support
  - Ensures rapid delivery of Sybase mobile and embedded solutions

Try It Yourself!

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- **Download FREE Sybase SQL Anywhere Studio Evaluation Copy**
  www.sybase.com/products/download.html
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- **Also, join the FREE Upcoming Beta Program**
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Questions???

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More connected.